UCAS / COMPUTER ENGINEERING

Robotics Programming

Week 1 | Course introduction



DSAI 4304
First Semester 2026–2027
Dr. Shadia Baroud

The question behind the course

CORE QUESTION

How does software turn sensor data into safe, goal-directed motion?

TODAY

Understand the course
See what ROS 2 connects
Know how to begin online

CHAT PROMPT

Name one sensor and one action you expect from a mobile robot.

Robot behavior runs in a feedback loop

A command changes the robot. New measurements show what happened.

01 Sense

Read lidar, odometry, or camera data.

02 Decide

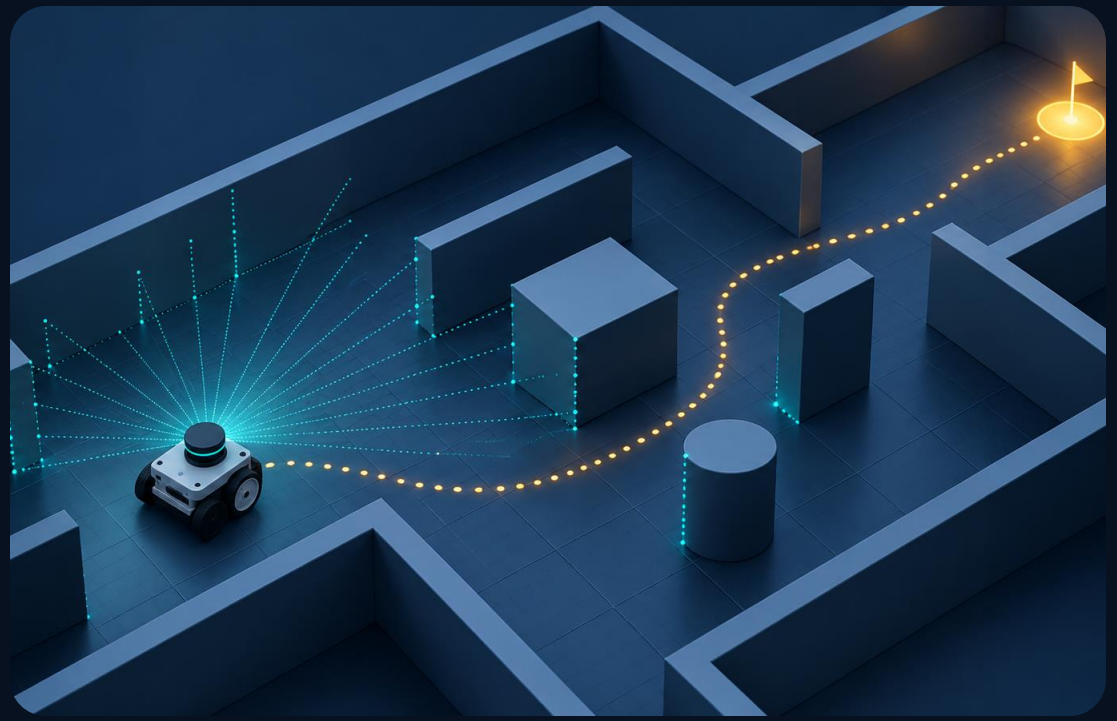
Estimate the situation and select a command.

03 Act

Send motion commands to the robot.

04 Check

Use the next measurements to adjust behavior.



A mobile robot gives us a shared example

We use a simulated differential-drive robot to connect each programming task to robot behavior.

MOTION

Two powered wheels set the robot’s forward speed and turn rate.

MEASUREMENTS

Lidar and odometry provide data that software can inspect and use.

WHY SIMULATION

Everyone can test the same system online. Physical hardware is not required.

ROS 2 connects the robot software

ROS 2 is middleware and tooling that lets independent programs exchange typed data and run as one robot system.

A useful distinction

ROS 2 is not a replacement for Ubuntu or Linux. It runs above the operating system and supplies libraries, communication patterns, and tools for robotics.

Nodes and communication interfaces

Each part has a different job in the ROS graph.

Concept	Purpose	Mobile-robot example
Node	A running program that performs part of the system’s work	Lidar driver or motion controller
Topic	A stream of typed messages	/scan readings or /cmd_vel commands
Service	A short request followed by a response	Ask a node to reset a setting
Action	A longer goal with feedback and cancellation	Navigate to a target pose

What you will learn to do

01 Build ROS 2 workspaces and packages, then write runnable Python and C++ nodes.

02 Design and test topics, services, actions, and custom message interfaces.

03 Configure a repeatable multi-node system with parameters and launch files.

04 Describe a mobile robot, interpret sensor data, and manage coordinate frames.

05 Apply mapping, localization, and navigation workflows in simulation.

Robotics foundations behind the code

Each concept helps answer a programming question.

Kinematics	How do wheel speeds affect motion?
Coordinate frames	Where was this measurement taken?
Sensor quality	How noisy or incomplete is the reading?
Feedback	Did the last command produce the intended motion?
Planning and localization	Where am I, and how can I reach the goal?

We introduce the theory needed to build and debug the software, then apply it in simulation.

Course progression

01

ROS 2 foundations

Workspaces, packages, nodes, Python and C++.

02

Communication

Topics, services, actions, parameters and launch.

03

Robot and sensing

Models, TF2, sensor data and simulation.

04

Autonomy

Planning, mapping, localization, SLAM and Nav2.

The midterm covers Weeks 1–7. The final is cumulative, with emphasis on integration, sensing, and navigation.

How the online course works

01	**Before class**	Short preparation and a readiness check.
02	**Live lesson**	Explanation, screen-shared coding, debugging, and questions.
03	**Guided lab**	Individual asynchronous work with simulation or supplied data.
04	**Submission**	Code, build/run evidence, a short explanation, and reproducible steps.

If local installation or connectivity is a barrier, use the posted recording, supplied data, or instructor-provided environment.

Software baseline and course access

PLATFORM

Ubuntu 24.04

ROS

ROS 2 Jazzy

SIMULATION

Gazebo Harmonic
RViz2

DEVELOPMENT TOOLS

Use Python 3 and C++. Build with colcon, track code with Git, and work in a code editor.

Simulation-first. No physical robot is needed. Follow the course setup guide for package names and version-specific instructions.

Course materials and announcements: moodle.ucas.edu.ps/course/view.php?id=894

Assessment plan

How your work contributes to the course grade

Component	Weight	Evidence
Weekly readiness checks	10%	Short online quiz or check-in on concepts and setup
ROS 2 programming lab portfolio	35%	Code, build/run evidence, brief explanation, reproducible instructions
Midterm examination	20%	Individual online exam covering Weeks 1–7
Final examination	35%	Cumulative online exam, with emphasis on integration, sensing, and navigation

Proposed weights are subject to approved UCAS requirements. Dates, duration, and permitted resources will be posted in Moodle.

Week 1 setup check: inspect the ROS graph

After following the course setup guide:

Run the supplied demo node.
Inspect active nodes and topics.
Record what publishes the data.
Explain the connection in two sentences.

ROS 2 CLI

```
ros2 node list
ros2 topic list
```

If your environment is not ready, use the instructor demo or recording and post the issue in Moodle.

Before our next session

PREPARE

Read Chapters 1–2 of the primary text and follow the course setup guide for ROS 2 Jazzy.

Carter, T., Sanders, R., & Farmer, M. (2025). Advanced Robotics Programming with ROS2 and Python: Mastering Navigation, Sensing, and Multi-Robot Systems in Real-World Applications.

EXIT TICKET

In one sentence, how does ROS 2 help two parts of a robot program communicate?

Moodle course: https://moodle.ucas.edu.ps/course/view.php?id=894